

Bayesian Analysis in Partially Identified Parametric and Nonparametric Models

Defense for Ph.D. in Statistics

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June 1, 2010

Outline

- 1 Introduction
- 2 Moment Inequality Model
 - Posterior
 - Moment and Model Selection
 - Numerical Example: Missing Data
- 3 Conditional Moment Restricted Model
 - Ill-Posed Inverse Problem
 - Proposed Bayesian Approach
- 4 Risk Control in Data Mining using Empirical Likelihood
- 5 Conclusion

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Introduction

Consider two types of moment conditions

① $Em(X, \theta) \geq 0$

② $E[m(X, g)|W] = 0$

- m is known up to the parameters
- Observe X, W .
- θ is finite dimensional, g is infinite dimensional.

Examples

Interval Regression

$$Y = X^T \theta_0 + \epsilon,$$

Instrumental variable Z : $E(Z\epsilon) = 0$.

$$Y \in [Y_1, Y_2], Z \geq 0 \Rightarrow$$

$$E(ZY_1) \leq E(ZX)^T \theta_0 \leq E(ZY_2)$$

$$Em(X, Z, Y_1, Y_2, \theta_0) = E \begin{pmatrix} ZX^T \theta_0 - ZY_1 \\ ZY_2 - ZX^T \theta_0 \end{pmatrix} \geq 0$$

Nonparametric Instrumental Variable Regression

$$Y = g_0(X) + \epsilon$$

- g_0 is nonparametric
- $E(\epsilon|X) \neq 0$, but $E(\epsilon|W) = 0$.
 - $\Rightarrow E[Y - g_0(X)|W] = 0$
 - $\Rightarrow E[m(X, Y, g_0)|W] = 0$

Moment Condition Based Bayesian Approach

- $p(\theta|Data) \propto p(\theta) \times \text{likelihood}.$

Construct the likelihood first.

- In practice,

What is known: moment conditions.

What is not known: the true likelihood function

- derive posterior from moment conditions

Zellner's BMOM Zellner (1994, 1995, 1996, 1997):

$$Y = X\beta + u$$

$$E(\beta|D) = (X'X)^{-1}X'Y, \text{Var}(\beta|D) = s^2(X'X)^{-1}.$$

Post-data density:

$$\max_f - \int f(\beta) \log f(\beta) d\beta$$

subject to $\int \beta^i f(\beta) d\beta = \mu_i$, where $\mu_i = E(\beta^i|D)$, $i = 1, 2$.

- Zellner's BMOM is based on $E[m(X, \theta)|X] = 0$.
- We consider moment conditions $E[m(X, \theta)|\theta] = 0$.

Bayesian Empirical Likelihood Lazar (2003)

$$\text{posterior} \propto p(\theta)L_{EL}(\theta)$$

Chernezukov and Hong (2003): if θ_0 is identified by $Em(X, \theta_0) = 0$,

$$\int \theta p_{EL}(\theta|Data)d\theta \rightarrow^P \theta$$

Bayesian GMM

$$Em(X, \theta) = 0$$

- Define $\bar{m}(\theta) = \frac{1}{n} \sum_{i=1}^n m(X_i, \theta)$.
- GMM criterion function: $\bar{m}(\theta)^T V^{-1} \bar{m}(\theta)$
- Bayesian GMM:

$$L_{GMM}(\theta) \propto \exp \left(-\frac{n}{2} \bar{m}(\theta)^T V^{-1} \bar{m}(\theta) \right)$$

- QUESTION: Does it have a likelihood interpretation?

Answer: Yes We Can! It is **Best Approximation to True**

Likelihood subject to moment conditions. Kim(2002)

“limited information likelihood”

Two Technical Problems

Identification $\arg \min_{\theta} Q(\theta) \neq \{\theta_0\}$

$$\text{e.g., } Q(\theta) = \inf_{\lambda > 0} \|Em(X, \theta) - \lambda\|^2, \\ \int E[m(X, \theta)|W]^2 dF(w)$$

Inverse problem Even if $\arg \min_{\theta} Q(\theta) = \{\theta_0\}$,

$$\hat{\theta} = \arg \min_{\theta} Q_n(\theta), \quad \hat{\theta} \not\rightarrow^P \theta_0$$

$$\text{regularization: } \hat{\theta}_{a_n} = \arg \min Q_n(\theta) + a_n^2 P(\theta)$$

Chen and Pouzo (2009)

Posterior:

$$\log p(\theta|D) \propto -n(Q_n(\theta) + a_n^2 P(\theta))$$

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Moment Inequality Model

$$Em(X, \theta_0) \geq 0$$

- Without further assumptions, the parameter is not identified.

$$\Omega = \{\theta \in \Theta : E_X m(X, \theta) \geq 0\}.$$

$$\Omega \neq \{\theta_0\}$$

- θ_0 : *partially identified* on Ω ;
 Ω : *identified region*.

Literature

Frequentist

- Inference: Chernozhukov, Hong and Tamer (2007), Beresteanu and Molinari (2008), Imbens and Manski (2004), Rosen (2008), Canay (2010), Bugni (2010), etc.
- Model Selection: Shi (2010), Andrews and Soares (2010)

Bayesian

- Moon and Schorfheid (2009), Bollinger and Hasselt (2009)

- $Em(X, \theta) \geq 0 \Leftrightarrow Em(X, \theta) = \lambda$, for some $\lambda \geq 0$

$$L(\theta, \lambda) \propto \exp \left(-\frac{n}{2} (\bar{m}(\theta) - \lambda)^T V^{-1} (\bar{m}(\theta) - \lambda) \right)$$

prior $p(\lambda) \sim \text{Exp}(\psi)$

- Assume $p(\theta, \lambda) = p(\theta)p(\lambda)$. Integrate out λ ,

$$p(\theta | \text{Data}) \propto p(\theta) \int_{[0, \infty)^p} L(\theta, \lambda) p(\lambda) d\lambda$$

- Frequentist property: posterior consistency

Technical Assumptions

- 1 The parameter space is compact.
- 2 The identified region has nonempty interior $int(\Omega)$.
- 3 $Em_j(X, \cdot) : \Theta \rightarrow \mathbb{R}$ is continuous for each component j .
- 4 The prior $p(\theta)$ is continuous and bounded away from zero on Ω .

Posterior Consistency

Theorem 2.1 (Posterior Consistency)

Assume the interior of Ω is not empty, then under regularity conditions:

- ① *For any small neighborhood of Ω , for some $a > 0$,*

$$P(\theta \notin \Omega^\delta | \text{Data}) = o_p(e^{-an})$$

- ② *$\forall$ open set $A \subset \text{int}(\Omega)$,*

$$\liminf_{n \rightarrow \infty} P(\theta \in A | \text{Data}) > 0$$

Moment and Model Selection

- Suppose we have p moment inequalities:

$$Em_i(X, \theta) \geq 0, \quad i = 1, \dots, p$$

where $\theta = (\theta_1, \dots, \theta_k) \in \Theta_1 \times \Theta_k$

- Moment selection: select moment inequalities;
Model selection: select parameter components θ .

Example: Dynamic Panel Data with Censoring

Similar to Andrews and Lu (2001)

$$y_{it} = z'_{it}\theta_t + \eta_i + v_{it}, \quad y_{it}^L \leq y_{it} \leq y_{it}^U$$

$$z_{it} = (X_{it}, w_{it}).$$

- $X_{it} = (x_{it-1}, \dots, x_{it-L})$. True L_0 is unknown.
- $Z_{it} = (w_{it}, \dots)$ Other variables not in the regression).

$$Ev_{it}Z_{it+1} = \dots = Ev_{it}Z_{iT} = 0$$

$$Z_{it} \geq 0, \text{ instrumental variable.}$$

- $E\eta_i = Ev_{it} = 0$

$$EZ_{it}(y_{it}^L - y_{it-1}^U) \leq EZ_{it}(z'_{it}\theta_t - z'_{it-1}\theta_{t-1}) \leq EZ_{it}(y_{it}^U - y_{it-1}^L),$$
$$Ey_{it}^L \leq Ez'_{it}\theta_t \leq Ey_{it}^U$$

- Moment Selection: select valid instruments Z_{it} .
- Model Selection: select different number of lags in X_{it} .

- Suppose we have $EM(X, \theta) \geq 0$.
- Select model and moments simultaneously $C_s = (M_s, \Theta_s)$.
- Let $\lambda = E_X M(X, \theta)$, where $M = (M_s, M_s^c)^T$.
- Limited Information Likelihood:

$$L(\theta_s, \lambda_s, \lambda_s^c) = \exp\left(-\frac{n}{2} G(\theta_s, \lambda_s, \lambda_s^c)\right)$$

where

$$G(\theta_s, \lambda_s, \lambda_s^c) = \left\{ \begin{pmatrix} \bar{M}_s(\theta_s) - \lambda_s \\ \bar{M}_s^c(\theta_s) - \lambda_s^c \end{pmatrix}^T V^{-1} \begin{pmatrix} \bar{M}_s(\theta_s) - \lambda_s \\ \bar{M}_s^c(\theta_s) - \lambda_s^c \end{pmatrix} \right\}$$

- $P(C_s | \text{Data}) \propto \int L(\theta_s, \lambda, C_s) p(\theta_s | C_s) p(\lambda_s | C_s) p(\lambda_s^c | C_s) d\theta_s d\lambda_s d\lambda_s^c \times p(C_s)$

Selecting Compatible Combinations

C_s is called compatible, if the identified region

$$\Omega = \{\theta_s \in \Theta_s : EM_s(X, \theta_s) \geq 0\} \neq \emptyset$$

Theorem 2.2

Under some regularity conditions,

- ① *If C_s is compatible ($\Omega \neq \emptyset$),*

$$\liminf_{n \rightarrow \infty} p(C_s | \text{Data}) > 0$$

- ② *If C_s is not compatible ($\Omega = \emptyset$), then for some $\alpha > 0$,*

$$p(C_s | \text{Data}) = o_p(e^{-\alpha n})p(C_s)$$

Consistency of Selection

- Liao and Jiang (2010) considered:
 - Simpler models are desirable
 - Contains as many moment inequalities as possible.
- It was shown that, if $C^* = \arg \max_C p(C|Data)$, asymptotically, C^* has the largest $\dim(M_s) - \dim(\Theta_s)$.
- Andrews and Lu (2001) assumed:

The true combination is the unique (M, Θ) , such that it has the maximal $\dim(M) - \dim(\Theta)$.

- Andrews and Lu (2001)'s assumption may fail in moment inequality models.
- In point identification case (moment selection in GMM), because of over-identification, it is reasonable that all the compatible moments combinations are nested.
- In partial identification case, there can be, however, more than one compatible combinations, some may not contain the true parameter.

- More reliable setting: put prior $p_{\theta,\lambda}(\theta, \lambda)$, $p(C_s)$.

$$EM(X, \theta) = \lambda \in \Lambda = \begin{cases} [0, \infty)^s \\ \mathbb{R}^{p-s} \end{cases}$$

Theorem 2.3

Let $\Omega(\Theta, \Lambda) = \{\theta \in \Theta : EM(X, \theta) \in \Lambda\}$. Under Assumptions,

$$\text{plim}_n p(C_s | X^n) = \text{Const} \times p(C_s) \int_{\Omega(\Theta, \Lambda)} p_{\theta,\lambda}(\theta_s, EM(X, \theta) | C_s) d\theta$$

- This is not a consistency result. The posterior is sensible to the choice of priors.

Numerical Example: Missing Data

- $Y = I(\text{employment})$
- $Z = I(\text{Nonmissing})$
- $P(Z = 0) = 0.33,$
- $P(Y = 1|\text{missing}) = 0.1, P(Y = 1|\text{Nonmissing}) = 0.7.$
- $\theta = P(Y = 1) = 0.5$
- Simulated 5000 data, where 1622 were missing.

$$\hat{P}(Y = 1|\text{Nonmissing}) = \frac{2355}{5000 - 1622} = 0.69$$

- “Missing at random” estimates $\hat{\theta} = 0.69$

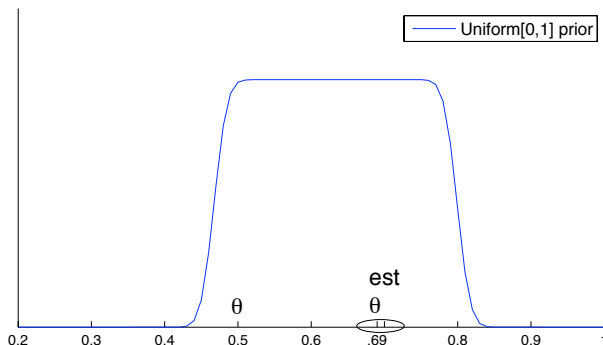
95% confidence interval $[0.68, 0.71]$.

Moment inequality approach

$$lower = P(Y = 1|Z = 1)P(Z = 1)$$

$$upper = P(Y = 1|Z = 1)P(Z = 1) + P(Z = 0)$$

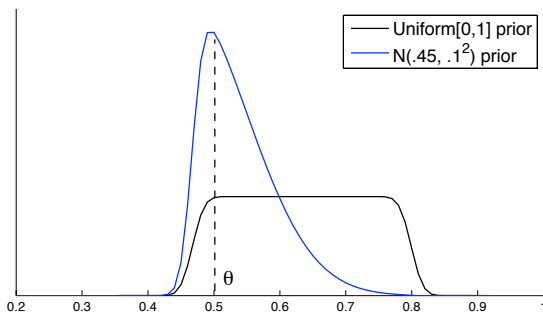
True ID region: $[0.47, 0.80]$. If $p(\theta) \sim \text{Uniform}[0, 1]$



Previous study (by a different method or with additional effort):
25 Y's are observed both for people with $Z = 1$ and $Z = 0$.

$$\hat{p} = 0.45, se = \sqrt{\frac{\hat{p}(1-\hat{p})}{25}} = .1$$

$$p(\theta) \sim N(.45, 0.1^2).$$



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Conditional Moment Restricted Model

$$E[\rho(X, g_0) | W] = 0$$

Examples:

- NPIV: $E(Y | W) = E[g_0(X) | W]$
- Single index model: $E(Y | W) = E[h_0(W^T \theta_0) | W]$
- Partially linear model:

$$E(Y | W_1, W_2) = E[h_0(W_1) + W_2^T \theta_0 | W_1, W_2]$$

- In NPIV, Newey and Powell (2003) showed that g_0 is identified iff $X|W$ is complete.
- The completeness condition is easy to fail:
(Severini and Tripathi 2006)

$X = W + U$: W and U are iid Uniform $[-\frac{1}{2}, \frac{1}{2}]$.

$E[g(X)|W] = 0$ iff

$g(x) = g(1+x) \forall x \in [-1, 0]$ and $\int_{-1}^0 g(x) dx = 0$.

- Sometimes instead of g_0 , we are interested in linear functional $h(g_0)$, e.g., $h(g_0) = \int_{-\pi}^{\pi} \sin(x)g_0(x)dx$.
- $h(g_0) = E[v(X)g_0(X)]$
Severini and Tripathi (2006): If $\exists m(W)$ s.t.

$$v(X) = E[m(W)|X]$$

$h(g_0)$ is identified.

- We do not assume g_0 is identified from $E[\rho(X, g_0)|W] = 0$.

$$\Omega = \{g \in \Theta : E[\rho(X, g_0)|W] = 0\}$$

is the identified region.

III-Posed Inverse Problem (partial identification)

- Define $T(g) = E[g(X)|W]$, and $\mu(w) = E(Y|W = w)$.

$$Tg_0 = \mu$$

- For any $g_1 \in L^2(X)$, define $[g_1] = \{g : Tg = Tg_1\}$. Define A on quotient space $L^2(X)/\mathcal{N}(T)$:

$$A[g] = Tg$$

But $A^{-1} : L^2(W) \rightarrow L^2(X)/\mathcal{N}(T)$ is not continuous.

$\Rightarrow$ CANNOT consistently estimate Ω by $\hat{A}^{-1}\hat{\mu}$.

Two Regularization Approaches

Approach I Assume $g_0 \in \Theta$, where Θ is compact
 $\Theta = \{g : \|g\| \leq B\}$ for some known $\|\cdot\|$ and $B > 0$.

- Result: If $T : \Theta \rightarrow L^2$, Θ is compact and T is compact, then T^{-1} is continuous.
- Newey and Powell (2003), Ai and Chen (2003), etc.

Approach II

- If $Q_n \Rightarrow Q \geq 0$ uniformly, $\arg \min Q(g) = g_0$,

$$\hat{g} = \arg \min Q_n(g) + a_n \|g\|^2, \quad a_n \rightarrow 0$$

under regularity conditions, $\hat{g} \rightarrow g_0$.

- Hall and Horowitz (2005), Chen and Pouzo (2009), Florens and Simoni (2009), etc.

Proposed Bayesian Approach

- Assume W is supported on $[0, 1]$.
- $E[\rho(X, g_0)|W] = 0$ implies

$$\underbrace{E \left(\rho(X, g_0) I(W \in [\frac{i-1}{k}, \frac{i}{k}]) \right)}_{m_i(D, g_0)} = 0, \quad i = 1, \dots, k$$

$$m = (m_1, \dots, m_k)^T.$$

- $Em(D, g_0) = 0$

- Suppose $g_0(x) \approx \sum_{i=1}^q b_i \phi_i(x)$, as $q \rightarrow \infty$
Put priors on $\{b_1, \dots, b_q\}$, fixing q .

$$g_q = \sum_{i=1}^q b_i \phi_i$$

- Likelihood is based on $Em(D, g) = 0$

$$p(g_q|D) \propto L_{GMM}(g_q)p(g_q)$$

Assumptions

- 1 Θ is compact under some norm $\|\cdot\|_H$.
- 2 (i) $P(W \in [\frac{j}{k}, \frac{j+1}{k}]) = O(k^{-1})$, $k = o(n^{2/5})$. (ii) $q = o(n)$.
- 3 $\max_{j \leq k} \sqrt{n} |\bar{m}_j(g) - Em_j(g, X)|$ is stochastic equicontinuous.
- 4 Continuity assumptions on ρ , $E(\rho|W)$, and $E(\rho(X, g_0)^2|W)$.
- 5 prior $p(g_q)$ has support around Ω .
- 6 $EY^2 < \infty$, $E\|m(D, g)\|^4 < B$ uniformly on Θ .

Posterior Consistency

Theorem 3.1

Let $\Omega = \{g \in \Theta : E(\rho(X, g)W)\}$. For any $U(\Omega)$,

$$P(g_q \in U(\Omega) | \text{Data}) \xrightarrow{P} 1$$

If in addition, $h(g_0)$ is identified, $h : \Theta \rightarrow \mathbb{R}$ is continuous, for any $\epsilon > 0$,

$$P(|h(g_q) - h(g_0)| < \epsilon | \text{Data}) \xrightarrow{P} 1$$

Relaxing the Compactness in NPIV

- Specify prior $\propto e^{-na_n^2 \|g\|^2}$,

limited Information Likelihood $\propto e^{-n\hat{Q}(g)}$

$$\log \text{posterior} = -n[\hat{Q}(g) + a_n^2 \|g\|^2]$$

- Require the prior variance $na_n^2 \rightarrow \infty$. Florens and Simoni (2009).

- Define $\|Tg - \mu\|^2 = \int \frac{(Tg(w) - \mu(w))^2}{E(\epsilon^2|w)} dF_W$.

$$\sup_{\|g\| \leq c} |\hat{Q}(g) - \|Tg - \mu\|^2| = o_p(1)$$

- Let the eigenvalues of T be $\lambda_1, \dots$, ordered such that $|\lambda_1| \geq |\lambda_2| \geq \dots > 0$, converging to zero.
- Define $g_0(x) = \sum_{i=1}^{\infty} g_i \phi_i(x)$,

$$\|g_0\| = \sum_{i=1}^{\infty} g_i^2$$

Assumptions

- 1 For some $\alpha > 0$, and and c ,

$$\sup_{\|g\| \leq c} |\hat{Q}(g) - \|Tg - \mu\|^2| = o_p(n^{-\alpha})$$

- 2 $a_n^2 \rightarrow 0$, and $na_n^2 \rightarrow \infty$

- 3 There exists $\{s_n\}_{n=1}^\infty \subset \mathbb{N}$, $s_n \rightarrow \infty$, such that

$$\sum_{j \geq s_n} g_j^2 = O(a_n^2 / \lambda_{s_n}^2) = o(1)$$

- 4 $n \succ q_n \succ \max\{n^{1-\alpha}, na_n^2 / \lambda_{s_n}^2, \lambda_{q_n}^{-2}\}$.

Posterior Consistency

Theorem 3.2

Assume g_0 is identified,

$$E[\|g_q - g_0\|^2 | \text{Data}] \xrightarrow{P} 0$$

- Prior variance $na_n^2 \rightarrow \infty$ for regularization.
Prior mean was set to zero.
- To incorporate prior knowledge of g_0 , e.g., convexity, monotonicity, use prior

$$\log p(g_q) \propto -na_n^2 \|g_q - g^*\|^2$$

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Risk Control in Data Mining

- We observe $(Y_1, X_1), \dots, (Y_n, X_n)$, where $Y_i \in \{0, 1\}$.
- classification rule $I(X^T \theta \geq 0)$
- risk $\Rightarrow$ moment condition

$$E|Y - I(X^T \theta)| = r$$

where (θ, r) is not point identified.

- Empirical likelihood based on $E|Y - I(X^T\theta)| = r$:

$$\log L_{EL}(\theta, r) = -\max_{\mu} \sum_{i=1}^n \log[1 + \mu(|Y_i - I(X_i^T\theta)| - r)]$$

- Joint posterior density: $p(\theta, r|Data) \propto \log L_{EL}(\theta, r)p(\theta, r)$.
- Applications:
 - 1 $p(r|\theta, Data)$, posterior of risk given the action
 - 2 $p(\theta|r \leq r_0, Data)$, for some fixed risk level of r_0 .
 - 3 $p(\theta \in M|r \leq r_0, Data)$, posterior of model M to control risk r_0 .

Main Results

Let $R(\theta) = E(|Y - I(X^T \theta \geq 0)| | \theta)$,

Theorem 4.1

Under some regularity conditions, for any $\epsilon > 0$,

$$P(r \in R(\theta) \pm \epsilon | \text{Data}) \xrightarrow{P} 1$$

Corollary 1

For any $\epsilon > 0$, $P(R(\theta) \leq r_0 + \epsilon | D, r \leq r_0) \xrightarrow{P} 1$, given that

$$P(r \leq r_0 | D) > 0.$$

German Credit Data: Empirical Application

- 1000 applications with credit rating $Y = \text{GOOD or BAD}$.
- 24 attributes: housing, credit history, employment, etc.
- The cost matrix

Table: Cost Matrix

		Classification	
		GOOD	BAD
Target Variable	GOOD	0	1
	BAD	5	0

- For identification in Horowitz (1992), define classification rule $C(X, \theta) = I(\theta_0 X_1 + \theta_1 + \sum_{i \geq 2} X_i \theta_i \psi_i > 0)$, where X_1 is continuous, $\theta_0 \in \{-1, +1\}$
- $\psi_i = 1$ if X_i is selected, otherwise $= 0$.
- loss $\rho(X, Y, \theta) = I(Y = \text{Good}, C = \text{Bad}) + 5I(Y = \text{Bad}, C = \text{Good})$.
- Empirical likelihood is based on $E(\rho(X, Y, \theta^\psi) | \theta, \psi) = r$.
- Prior $\theta^\psi | \psi \sim N(0, 10I)$, $r \sim \text{Uniform}$, $\psi_i \sim \text{Bino}(1, 0.1)$.

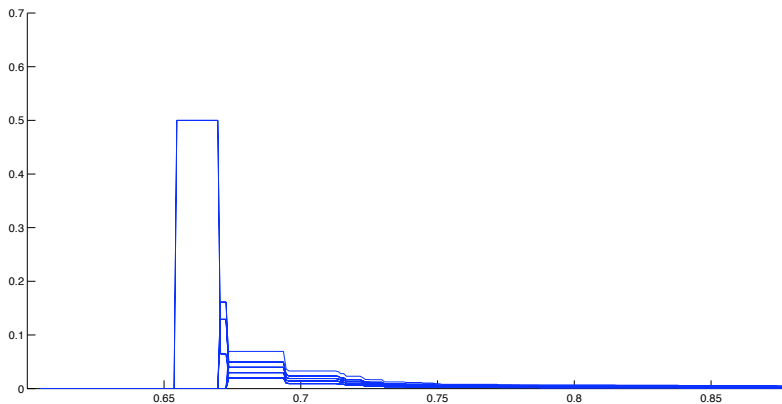


Figure: Estimated $P(\psi|r \leq r_0, \text{Data})$ versus r_0

Model	Variables	Posterior Probability
M_1	Duration of Credit Credit History Credit Amount Present Employment Since Real Estate Property Age Num. of Existing Credits at Bank Num. of People Being Liable Other Debtors/ Guarantors Credit Purpose	0.5
M_2	M_1 / Credit Purpose Telephone	0.5

- Partition data into Training (2/3) and Validation (1/3).

$$\hat{R} = \frac{1}{n_v} \sum_{i \in \text{Validation}} E[\rho(Y_i, X_i, \theta) | r \leq r_0, \text{Model}, \text{TrainingData}]$$

Compare with r_0 to see if r_0 can be practically achieved by selected models.

- $\theta^{\text{Model}} = \arg \min \frac{1}{n_c} \sum_{\text{Training}} \rho(Y_i, X_i, \theta^{\text{Model}})$

$$\min ER = \frac{1}{n_v} \sum_{i \in \text{Validation}} \rho(Y_i, X_i, \theta^{\text{Model}})$$

Compare with $\frac{1}{n_v} \sum_{i \in \text{Validation}} \rho(Y_i, X_i, \theta)$, to see whether model selection indeed leads to better prediction

Model	$\hat{R}$	r_0	min- <i>ER</i>	<i>ER-ALL</i>
M_1	0.679	0.680	0.682	0.730
M_2	0.659		0.679	

Table: Classification Accuracy

Model	Type I	Type II	Overall
M_1	0.195	0.916	0.426
M_2	0.478	0.841	0.595

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Conclusions

- Loss of Identification exists commonly in statistical models incomplete data, incompleteness of the distribution.
- Bayesian approach can incorporate prior information
- If the model is characterized by just moment conditions, moment-condition-based likelihoods are applied
- I studied the frequentist properties of the posterior:

Posterior consistency

- Estimation
- Model selection

Future Works

- Moment-condition-based Bayesian analysis has not been well studied
 - optimality in a frequentist sense
 - finite sample interpretation and properties
 - seek for pure Bayesian approach
- Moment inequality Applications
- Thank you very much!